

EE 210L: Digital Circuits and Logic Design Lab

Lab 11: Counters and Memory on the FPGA

Name: _____ Date: _____

Partner(s): _____ Section: _____

Do not start Part 5.2 until the LED in Part 5.1 is blinking. Everything below depends on a working clock.

Part 5.1: Verify the Clock

Divider width used: _____ bits

Time for ten blinks: _____ s Period per blink: _____ s

Clock frequency computed from your measurement: _____ MHz Expected: 125 MHz

Show your divider calculation:

Part 5.2: Counter on Hardware

Time for a full 0→15 cycle: _____ s Predicted from your divider: _____ s

Does btn[0] reset the count? _____

Resource	Used	% Utilization
LUTs		
Flip-flops		

Part 5.3: Ring Counter

Describe what the LEDs do, and how it differs from the binary counter:

Does this match the 74LS194 behavior from Lab 7? _____

Part 5.4: Button Without Debouncing

Trial	Presses made	Count reached	Extra counts
1	10		
2	10		
3	10		
4	10		

Part 5.5: Button With Debouncing

Trial	Presses made	Count reached	Extra counts
1	10		
2	10		
3	10		
4	10		

What did the debouncer fix, and roughly how many extra counts was bounce producing per press?

Why is the two-flip-flop synchronizer needed even though the debouncer already filters the input?

Part 5.6: ROM Addressed by the Counter

List the 16 values you stored.

Addr	0	1	2	3	4	5	6	7
Value								
Addr	8	9	10	11	12	13	14	15
Value								

Did the LEDs display your pattern rather than a binary count? _____

Resource	Used
LUTs	
Block RAM (BRAM tiles)	

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Was the ROM built from LUTs or block RAM? _____

Part 5.7: RAM Write and Read Back

Address	Value written	Value read back	Match?

Part 5.8: Forcing Block RAM

Implementation	LUTs used	BRAM tiles used
Default (tool's choice)		
ram_style = "block"		

Which is the better fit for a 16×4 memory, and why did the tool choose what it did by default?

Discussion and Conclusions

Address at least: why you divide a clock with an enable rather than by clocking logic from a divided signal, what simulation could not have shown you about the button, and when a design should use block RAM instead of distributed RAM.
